

Title

CI/CD Sentinel: Event-Driven Observability Platform for CI Pipelines

Problem Statement

Modern software projects rely on CI pipelines like GitHub Actions to automate build and testing processes. However, pipeline status and reliability metrics are scattered across different dashboards, reducing visibility into system stability. Developers often detect failures late due to manual monitoring practices. Lack of centralized insights makes it difficult to track build success rate, failure trends, and recovery time. This creates operational risk and reduces software reliability.

Proposed Solution

CI/CD Sentinel provides a centralized platform that collects CI pipeline events using secure webhooks and processes them through an event-driven architecture. The system computes reliability metrics such as success rate, build duration, and failure frequency. Automated validation ensures reproducible builds, dependency security checks, and structured logging. The platform enables proactive detection of failures before production deployment. The design emphasizes automation, reproducibility, and disciplined engineering workflows.

Module Description

Webhook API

Provides a centralized interface for transport authorities to manage routes, stops, schedules, buses, and drivers. Enables creation and monitoring of trips with structured control over fleet operations. Displays live bus locations on map for operational awareness. Ensures administrative visibility and decision support.

Event Queue

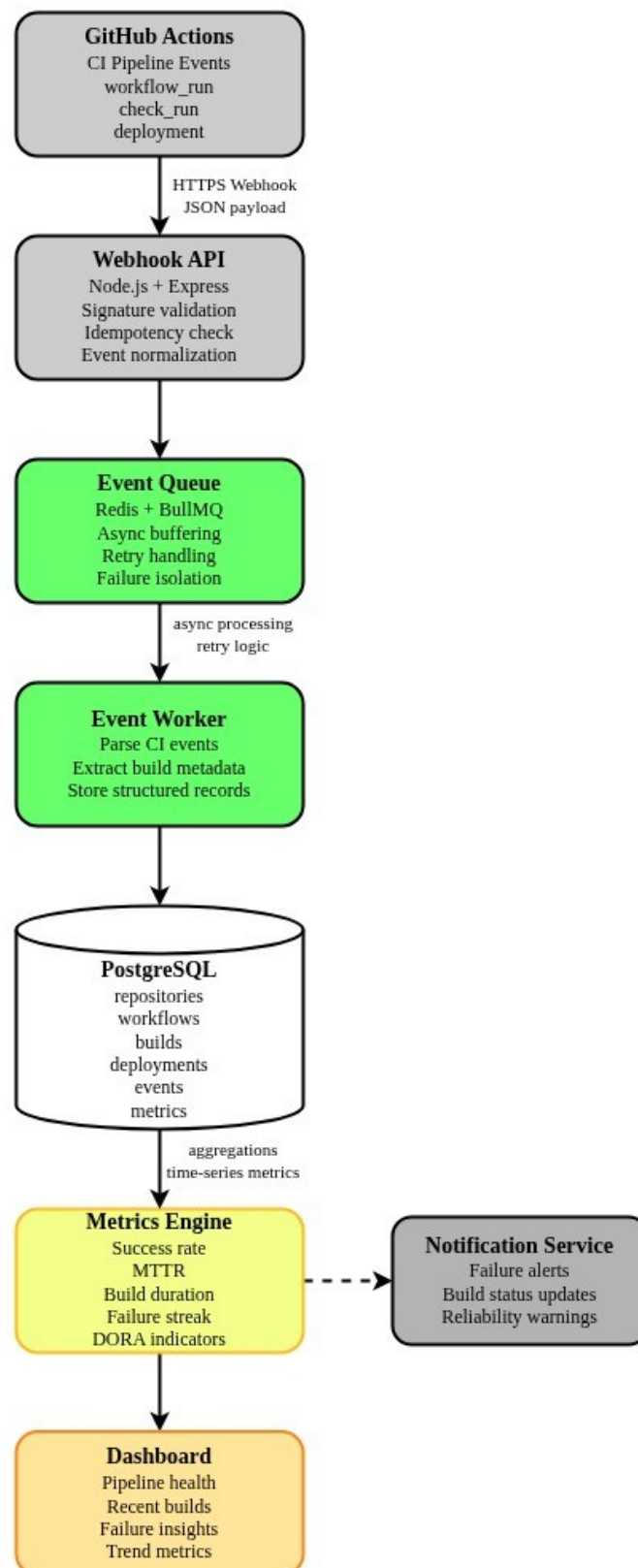
Buffers incoming events to handle bursts in CI activity. Enables asynchronous processing and improves system reliability.

Event Queue

Processes pipeline event data and stores structured build information. Normalizes CI data for consistent analytics computation.

Architecture Diagram

CI/CD Sentinel — Architecture Diagram



Metrics Engine

Calculates reliability indicators such as success rate, MTTR, failure streak, and build duration trends. Provides measurable insights into pipeline stability.

Dashboard & Alerts

Displays pipeline health indicators and recent build results. Sends notifications when failures or abnormal patterns are detected.

USP (Unique Selling Proposition)

- Centralized CI pipeline observability
- Event-driven architecture for reliable processing
- Automated reliability metric computation
- Reproducible build validation strategy
- Integrated dependency security checks
- Scalable design for future multi-provider CI support

Tech Stack

Backend: Node.js, TypeScript

Database: PostgreSQL

CI Integration: GitHub Actions Webhooks

Queue: Redis + BullMQ

Containerization: Docker

Logging: Structured logging (Pino)

Deployment: Cloud platform (Render/AWS)